NAME: _____ Geometry (FL B.E.S.T.)

12.1 Probability and Sample Spaces

The **sample space** S is the set of all possible outcomes of an experiment; an **event** E is any subset of those outcomes. For equally likely outcomes, $P(E) = \dfrac{n(E)}{n(S)}$ — the count of favorable outcomes over the total count. Two techniques make counting outcomes manageable: listing/organizing the full sample space (a grid for two dice, a tree for sequential flips), and the **complement rule** $P(E') = 1 - P(E)$, which is often far faster than counting the event directly — especially for "at least one" questions. All probability answers in this course should be exact fractions in lowest terms.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. Fair die. Find $P(\text{odd})$.

2. Fair die. Find $P(\text{at least } 5)$.

3. Bag: 4 red, 6 blue. Find $P(\text{red})$.

4. Bag: 4 red, 6 blue. Find $P(\text{not blue})$.

5. Two fair coins. Find $P(\text{two heads})$.

6. Spinner 1–10. Find $P(\text{multiple of } 3)$.

7. Find $P(\text{ace})$ from a 52-card deck.

8. List the sample space S for one flip of a fair coin.

9. What is the range of possible probability values?

10. Find $P(\text{die shows 1 through 6})$.

11. Find $P(\text{die shows 7})$.

12. Three fair coins. Find $P(\text{exactly two heads})$.

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13. $P(E) = 0.35$. Find $P(E')$.

14. Two fair dice. Find $P(\text{sum} = 8)$.

15. Two fair dice are rolled. Find $P(\text{sum is a prime number})$ — remember $2, 3, 5, 7, 11$ are the possible prime sums.

✓ Answer Key

1. $\frac{1}{2}$
2. $\frac{1}{3}$
3. $\frac{2}{5}$
4. $\frac{2}{5}$
5. $\frac{1}{4}$
6. $\frac{3}{10}$
7. $\frac{1}{13}$
8. $\{H, T\}$
9. 0 to 1, inclusive
10. 1
11. 0
12. $\frac{3}{8}$
13. 0.65
14. $\frac{5}{36}$
15. $\frac{5}{12}$